

BS Software Engineering Suggested Four Year Plan
2017 – 2018 Catalog

6/1/2017

CS 1336 is required of students with no prior programming experience
Not For Credit in ECS C or better needed

Freshman

Description	Course No
Introduction to CS & SE	CS 1200
Introduction to Engr & CS	ECS 1100
Calculus I	Math 2413 or 2417
Computer Science I	CS 1337
6 hours CORE: Choices below	
Freshman Seminar	UNIV 1010

UNIV 1010

ECS 1100

*Math 2413

*Math 2417

CS 1200

**CS 1337

CORE

CORE

Freshman

Description	Course No
Computer Science II	CS 2336
Discrete Mathematics I	CS 2305
Calculus II	Math 2414 or 2419
Mechanics and Heat	Phys 2325
Mechanics and Heat Lab	Phys 2125
3 hours CORE: Choices below	

Phys 2325
Phys 2125

Math 2414

Math 2419

Math 2312 (C or better)
or 75% on ALEKS

**CS 2305

**CS 2336

CORE

Sophomore

Description	Course No
Mathematics Foundations of SE	SE 3306
Computer Architecture	SE 3340
Electricity and Magnetism	Phys 2326
Electricity and Magnetism Lab	Phys 2126
6 hours CORE: Choices below	

Phys 2326
Phys 2126

SE 3340

SE 3306

CORE

CORE

Sophomore

Description	Course No
Probability & Statistics in CS	SE 3341
Algorithms Analysis & Data Struct	SE 3345
C/C++ Prog'ing in a UNIX Env.	SE 3377
Linear Algebra	Math 2418
3 hours CORE: Choices below	

Math 2418

SE 3341

SE 3345

SE 3377

CORE

At this point, you should choose a domain.

Junior

Description	Course No
Software Engineering	SE 3354
Operating Systems	SE 4348
SE Guided Elective	SE 43XX
Prof & Tech Communication	ECS 3390
3 hours CORE: Choices below	

SE 43xx
Guided Elective
See Advisor

SE 4348

SE 3354

ECS 3390

CORE

**CS 2305 &
**CS 2336

Junior
Rhet 1302

Junior

Description	Course No
Software Project Planning & Mgmt.	SE 4381
Software Architecture & Design	SE 4352
3 hours SE Guided Elective	SE 43XX
Social Issues & Ethics in ECS	ECS 3361
Database Systems	SE 4347
Prof. Responsibility in CS & SE	SE/CS 3162

SE 4347

SE 4381

SE 4352

SE 3162

ECS 3361

SE 43xx Guided
Elective
See Advisor

At least two of
these 4 complete
to begin Project.

Senior

Description	Course No
Software Testing, Verif, Valid, QA	SE 4367
Requirements Engineering	SE 4351
3 hours SE Guided Elective	SE/CS 43XX
3 hours Free Elective	See Advisor
3 hours Science Elective	See Advisor

SE 43xx Guided
Elective
See Advisor

SE 4351

SE 4367

Science
Elective
3 Hours

Free
Elective

Senior

Description	Course No
Software Engineering Project	SE 4485
3 hours SE Guided Elective	SE 43XX
5 hours Free Elective	See Advisor
Core Curriculum Assessment	UNIV 2020

SE 43xx Guided
Elective
See Advisor

Free
Elective (2)

SE 4485

Free
Elective

UNIV 2020

*A grade of C- or better is needed in Math 2413 or Math 2417 to move forward.
**A grade of C or better is required for CS 1336, CS 1337, CS 2305, and CS 2336.

Core Choices

Rhet 1302	3 hours From:	3 hours From:	6 hours From:
	Ahst 1303, 1304, 2331	Ams 2300	Hist 1301
	Arts1301	Ams 2341	Hist 1302
	Danc 1310	Huma 1301	Hist 2301
	Film 2332	Lit 2331	Hist 2330
	Musi 1306	Phil 1301	Hist 2332
	Thea 1310	Phil 2316, 2317	

Courses may be taken in any order, as long as prerequisites are met.

Domains and Guided Electives (12hrs)

Students should complete three courses from one of the application domains. See the document entitled "Application Domains." Further guided electives can be selected from the CS/SE 43xx level classes.

LEGEND

Prereq Critical Path

Pre-requisite

Pre- or Co-requisite